08D — Deep-Dive Feature Audit: Outside-Sect World & Ecology (Part 4 of 6)

Wilderness Ecology, Roaming Spirit-Beast Packs, Menagerie Breeding (`menagerieTick` Bug!), Orphaned `reservePets`, Beast Symbiosis & Wild Resource Gathering

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Core Game Philosophy**: **100% Emergent, Autonomous Xianxia Sandbox**. Spirit beasts must inhabit logical biomes, navigate real terrain without walking onto cliffs or deep water, breed naturally inside the Sect Menagerie, pass down to apprentices or `BEAST_SOVEREIGN` disciples when their master dies, and merge with symbiotic warriors in life-or-death battles!
- **Master Roadmap for Category 08 (`08A–08F`)**:
  - `08A`: Macro Geopolitics, 19 Rival Sects, Diplomacy, Vassals, 6 Resource Nodes & Siege Warfare
  - `08B`: Expeditions, Caravans, Midnight Black Market, Tavern Bounties, Espionage & `v2.6.0` Living Chronicle
  - `08C`: Secret Realms, 43-Floor Nine-Dao Pagoda, 4 Sanctums, Puzzle Gates, 4 Micro-Locations & Immortal Tombs
  - **`08D` (This Document)**: **Wilderness Ecology, Roaming Spirit-Beast Packs, Taming/Pets/Menagerie/Symbiosis & Wild Resource Gathering (`wildlife.js`, `wild_resources.js`, `data_beasts.js`, `cultivation.js`, `sim.js`)**
  - `08E`: Southern Spirit River Fishing, Spirit Cuisine, Underwater Chests, Wandering Figures, Visitors & Legendary Clue Quests
  - `08F`: Grand Finale — Holistic Overview of the Entire Outside-Sect World & Ecology

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1. [WLD-MENAG-01] Critical Engine Bug in js/sim.js:186: CULT.menagerieTick(W) Is Called ONLY Once Before W.sect Exists — Never Running Again on Any Day or Month of Gameplay!

- **Exact Code Proof (`js/sim.js:185–192` vs `480–545`)**:

```javascript
// sim.js line 185:
const W = { _founding: true, ..., npcs: [], ... }; // <-- W.sect IS STILL UNDEFINED HERE!
if (g.CULT && g.CULT.menagerieTick) g.CULT.menagerieTick(W); // <-- ABORTS IMMEDIATELY!
...
// sim.js line 192 (6 lines later!):
W.sect = { name: ..., ... };
```

- **Verified Consequence**:
  - Inspecting the entire `sim.js` file (`lines 1–680`) confirms `CULT.menagerieTick(W)` is **never called in `SIM.dailyUpdate` or `SIM.monthly`**—its sole call site in `sim.js` is line `186`, before `W.sect` is even initialized!
  - Consequently, autonomous Menagerie care, breeding progress, and `beast_menagerie` facility ticks inside `CULT.menagerieTick` **never execute during live simulation**!
- **Exact Fix**: Remove the premature line-186 call before `W.sect` exists, and invoke `if (g.CULT && g.CULT.menagerieTick) g.CULT.menagerieTick(W);` inside `SIM.dailyUpdate(W)` (or `SIM.monthly(W)` per its intended cadence) after `W.sect` is active!

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2. [WLD-PET-02] Orphaned W.sect.reservePets Dead-End (sim.js:495–501), Petless BEAST_SOVEREIGN Trap, & Dormant BEAST_SYMBIOSIS!

1. **Loyal Spirit Beasts Rot Forever in `W.sect.reservePets` After Their Master Dies (`js/sim.js:495–501`)**:

- When an Elder with bonded Spirit Beasts (`n.pets`) dies, `SIM.dailyUpdate` (`lines 495–501`) moves all of `n.pets` into `W.sect.reservePets` (`setting orphan.active = false`).
- **Broken Loop**: **Zero autonomous NPCs ever adopt from `W.sect.reservePets`!** Neither the deceased Elder's personal disciple (`apprentice`), spouse (`คู่ครอง`), the `beastmaster` officer, nor petless `BEAST_SOVEREIGN` disciples ever bond with orphaned beasts in `W.sect.reservePets` (and `SIM.retire` on line `347` forgets `n.pets` completely!).
- **Emergent Wuxia Fix**:
  - First priority on Master's death (`AI.kill`): Let a loyal personal disciple (`apprentice`) or spouse on a compatible path **inherit the deceased Elder's Guardian Spirit Beast** (`preserving high bond + logging a moving Chronicle entry: "อสูรพิทักษ์ของผู้อาวุโสยอมรับศิษย์สืบทอดเป็นนายคนใหม่"`)!
  - Second priority: Let `beastmaster` Jurisdiction (`runOfficerJurisdictionSingle(W, 'beastmaster')`) and `BEAST_SOVEREIGN` / `PRIMORDIAL_SHIFTER` / `SPIRIT_HUSBANDRY` disciples (`with pets.length < maxPets`) adopt compatible beasts from `W.sect.reservePets`!

2. **Petless `BEAST_SOVEREIGN` Disciples Lack an `'acquire_spirit_beast'` Cognition Goal (`01D`)**:
   - Newly awakened `BEAST_SOVEREIGN` disciples start with `n.pets = []` (`losing 100% of their path's signature pet stats & coordinated strikes!`), yet `COG` never prioritizes hunting/taming a wild beast (`wildlife.js`) or adopting from the Menagerie!
   - Also wire autonomous **Sovereign Bloodline Awakening** (using surplus `CAT_PRIMORDIAL_*` catalysts via `beastmaster`) and **`CULT.BEAST_SYMBIOSIS.activatePossession`** mid-combat when a `SYMBIOTIC_BEAST_WARRIOR` (`or stage >= 4 BEAST_SOVEREIGN`) drops below `65%` HP in a `serious: true` battle!

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3. `[WLD-BIO-03]` Taxonomy & Biome Bugs in `data/data_beasts.js` + Wild Beasts Walking Into Cliffs & Deep Water (`js/wildlife.js`)!

1. **Ocean Leviathans Tagged `"urban"` & `"forest"`, and Venom/Blood Beasts Tagged `"WIND"` (`data/data_beasts.js`)**:
   - Inspect `data/data_beasts.js` verbatim:

    - **BEAST_ABYSSAL_DRAGON (มังกรสมุทรเก้าเศียรบรรพกาล — Nine-Headed Ocean Dragon, Tier 7)**: `"terrain": "urban"` (**city terrain!** Because `ecology_expansion.js` Test Group 10 checks `sp.terrain !== 'water'`, tagging an Ocean Dragon `"urban"` bypassed the aquatic filter!). Change to `"terrain": "water", "archetype": "AQUATIC"` (keeping Test Group 10 100% green while moving it to water expeditions!).
    - **BEAST_ABYSSAL_LEVIATHAN_TURTLE (เต่ามังกรสมุทรดึกดำบรรพ์, Tier 5)**: `"terrain": "forest"` $\to$ Change to `"terrain": "water", "archetype": "AQUATIC"`.
    - **BEAST_THUNDER_DRAKE_LIZARD (กิ้งก่ามังกรสายฟ้าทะเลทราย, Tier 4)**: `"terrain": "forest", "element": "FIRE"` $\to$ Change to `"terrain": "desert", "element": "LIGHTNING"` to match its own name & `THUNDER_LIZARD_SCALE` drop!
    - **Element Mismatches**: `BEAST_EMERALD_PYTHON` (เหลือมมรกตเจ็ดพิษ) is `"WIND"` ($\to$ `"WOOD_LIFE"` / `"YIN_SHADOW"`), and `BEAST_BLOOD_FIEND_HOUND_4` (อสูรหมาป่าโลหิตอาฆาต) + `BEAST_MYTHIC_TAOTIE_SHADOW` are `"WIND"` ($\to$ `"BLOOD_DEMON"` / `"YIN_SHADOW"`).
2. **Wild Beasts Drift Onto Impassable Cliffs, Walls & Deep River Tiles (`js/wildlife.js`)**:
    - During `WILD.tick` wandering updates, verify `!MAP.blocked(W.map, nx, ny)` (`M.block[idx] === 0`) **and** `!MAP.isInRiverBelt` water tiles before committing `(b.x, b.y)` so wild beasts never get stuck inside cliff/water/rock tiles where melee hunters cannot stand and `checkLineOfSight` blocks ranged attacks!

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4. `[WLD-RES-04]` Wrong Office IDs (`'medicine'`/`'herbs'`) in `js/wild_resources.js`, Telepathic Node Radar, & Missing Beast Crop-Grazing Loop!

1. **Typo in `AI.ACTS.gatherWildResource.score` (`js/wild_resources.js`)**:
    - Checks `n.post === 'medicine' || n.post === 'herbs'` (**neither ID exists in `SECT.POST_IDS`**—canonical IDs are **`'healer'`** and **`'herbal'`**!). Fix to `['healer', 'herbal', 'wildwarden'].includes(n.post)`!
2. **Fog-of-War Discovery for Exterior Wild Nodes & Legendary Beast Sightings (`02A`)**:
    - Instead of every indoor disciple telepathically knowing the instant a rare T4 herb or Legendary Beast spawns outside the walls, mark newly spawned distant nodes `discovered = false` until spotted by a `guard`/`tower` watchman, `wildwarden` patrol, or passing forager (`or sensed immediately by`

`high-perceptiveness / SPIRIT_HUSBANDRY / WANDERING_DAO elders`), who then posts the sighting at `gate` / `mission_hall`!

3. **Connect Wild Beast Herb-Grazing With `wildwarden` & `03F` Garden-Trap Arts (`ART190_*_072`)**:
    - If wild boars/stags roam unchecked near exterior/open `herb_garden` / `grove` without an active `wildwarden` patrol or `ART190_*_072` Talisman Garden Trap (`03F`), let them occasionally graze `1` common herb (`while triggering garden traps damages/captures the beast for free meat/pelts!`).

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5. Mandatory Verification Suite Additions (`08D`)

1. **Live `CULT.menagerieTick` Execution Test**: Run `SIM.dailyUpdate(W)` on a world with 2 compatible beasts in the Menagerie and assert `menagerieTick` advances breeding/care state.
2. **Orphan `reservePets` Inheritance & Adoption Test**: Kill an Elder owning a Tier-4 pet; assert their `apprentice` or a petless `BEAST_SOVEREIGN` disciple adopts the pet from `reservePets` rather than leaving it permanently inactive.
3. **Wilderness Walkability Invariant Test**: Simulate `1,440` ticks of `WILD.tick(W)` and assert **every** living wild beast satisfies `MAP.blocked(W.map, b.x, b.y) === false`.